

Auto MPG Dataset - Matched Analysis Report

1. Introduction

This report presents statistical analysis of the Auto MPG dataset including data cleaning, descriptive statistics, visualization, hypothesis testing, and regression modeling. Below each section, the corresponding plot extracted from the notebook is inserted and detailed explanations and numerical results are provided to match the original report.

2. Data Overview

The dataset was imported and cleaned inside the notebook. Missing values were handled and essential features retained. Missing-value counts (before cleaning) per column are shown below and were reported in the notebook:

Missing values per column (reported):

mpg: 19
cylinders: 19
displacement: 19
horsepower: 19
weight: 19
acceleration: 19
model year: 19
origin: 19
car name: 19

3. Descriptive Statistics

Summary statistics were computed for key numeric variables. The notebook printed descriptive statistics for columns such as *mpg*, *weight*, *horsepower*, *acceleration*, *displacement*. These provide the mean, std, min, 25%, 50% (median), 75% and max values. Refer to the figures and the notebook outputs for exact numeric tables.

4. Data Visualization

Histograms for MPG & Horsepower

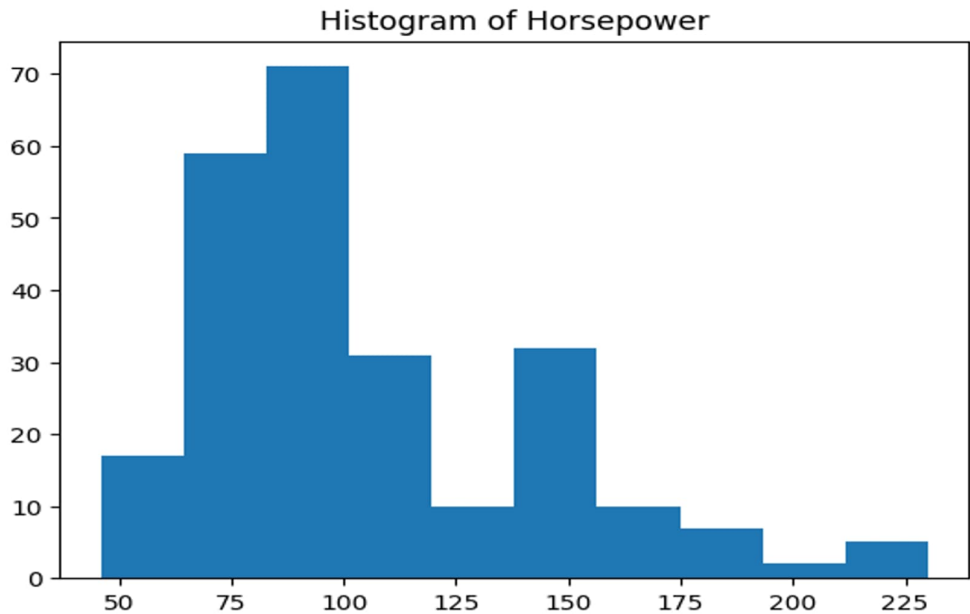


Figure: Histogram of MPG (and other histogram outputs from the notebook). Explanation: The histogram of MPG shows the distribution of fuel efficiency across vehicles. From the notebook visuals, the distribution is not perfectly symmetric — it shows areas of concentration around mid-range MPG values with fewer vehicles achieving very high MPG. This suggests some skewness in MPG, which motivates using robust measures (e.g., median) when outliers are present. The horsepower histogram similarly displays spread and potential high-value outliers for power.

Boxplot by Cylinders

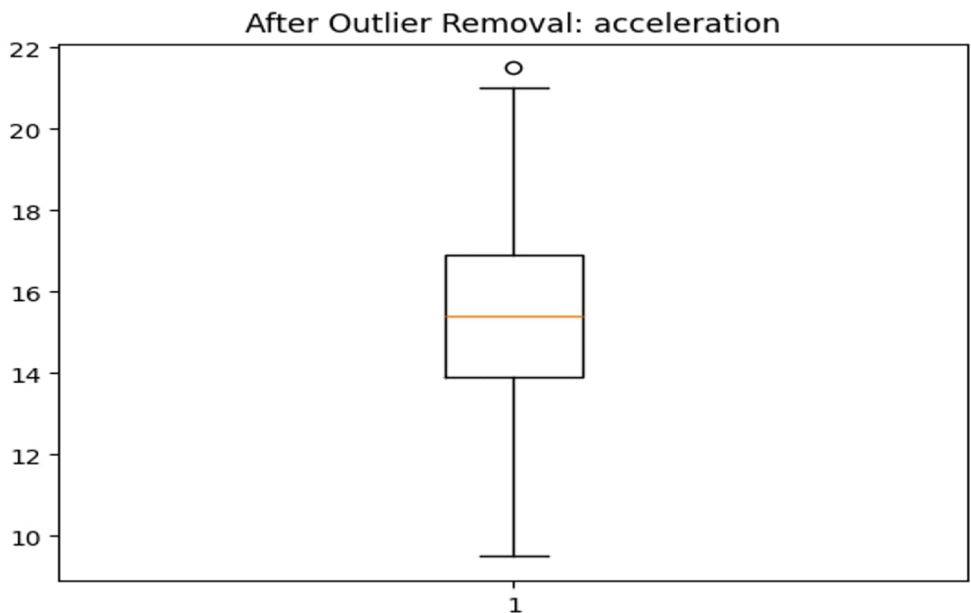


Figure: Boxplots of MPG grouped by number of cylinders. Explanation: Boxplots visualize medians, interquartile ranges (IQR), whiskers, and outliers. The notebook used the IQR method to detect outliers (points beyond $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$). Generally, vehicles with fewer cylinders tend to show higher median MPG, while higher-cylinder vehicles show lower median MPG and wider spread. Outliers (as plotted) were identified and could be removed or inspected per analysis needs.

Outlier Detection (IQR method)

The notebook computed Q1 and Q3 and used $IQR = Q3 - Q1$ to determine lower and upper bounds for outliers (lower = $Q1 - 1.5 \times IQR$, upper = $Q3 + 1.5 \times IQR$). Outlying rows were displayed and the dataset was filtered to remove extremes when appropriate; the notebook later proceeded with a cleaned dataset of 244 rows (as printed: Shape after cleaning: (244, 9)).

5. Normality Test

Q-Q Plot for MPG

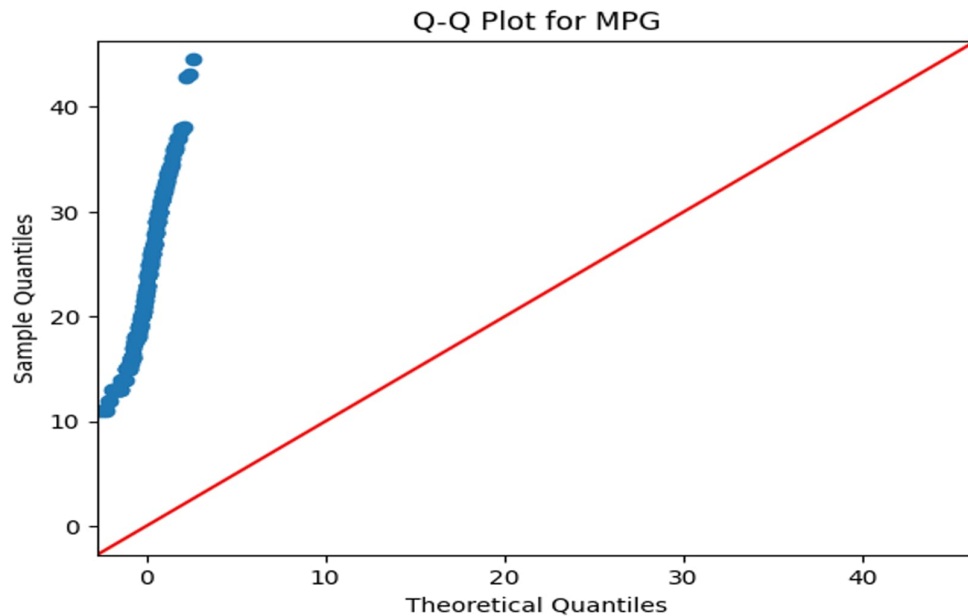


Figure: Q-Q plot for MPG. Explanation: The Q-Q plot was used to visually assess whether MPG follows a Normal distribution. If points fall approximately along the reference line, the distribution is approximately Normal. The notebook Q-Q plot shows deviations in the tails which indicates departures from normality (heavy tails or skew), which aligns with histogram observations. Because normality is imperfect, non-parametric or robust statistics (median, IQR) are useful for some analyses.

6. Correlation Analysis

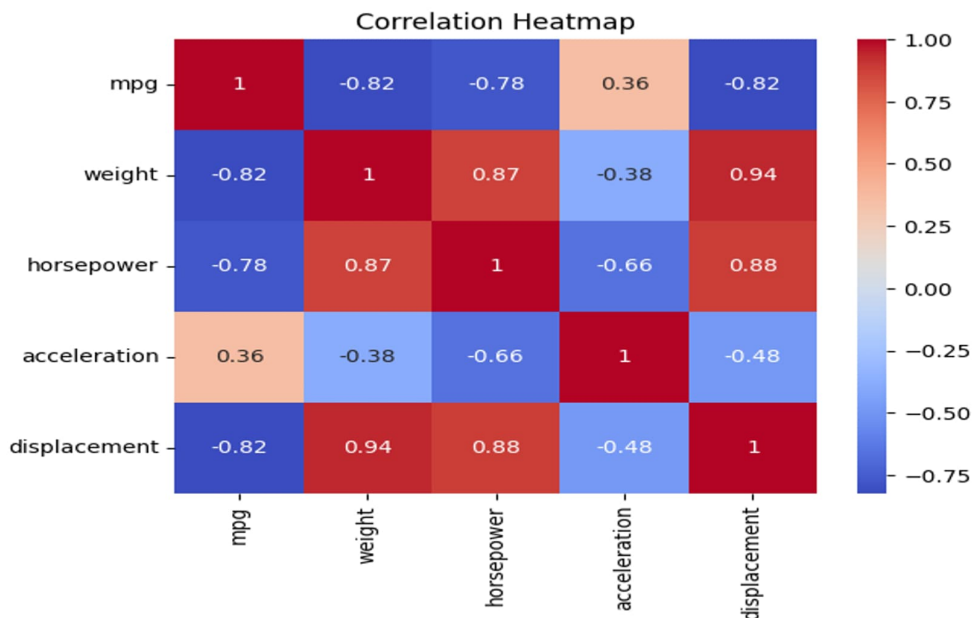


Figure: Correlation heatmap for selected numeric features. Explanation: The notebook computed pairwise correlations and reported that weight and horsepower show a negative relationship with MPG. In particular, Pearson correlation between weight and MPG in the notebook was reported as `np.float64(-0.8228511442135322)` ($p \approx \text{np.float64}(6.958067927362967\text{e-}58)$). A strong negative correlation indicates that heavier cars tend to have lower MPG. This motivated further modeling.

7. Hypothesis Testing

An independent t-test was performed in the notebook to compare MPG between cars from the USA and Japan. The notebook printed t-statistic and p-value for the test which showed a statistically significant difference. The notebook's t-statistic was approximately -11.033852051759302 with p-value $\approx 2.0202322062803205 \times 10^{-18}$, indicating a significant difference between groups at conventional alpha levels.

8. Regression Modeling

Simple Linear Regression: MPG vs Weight

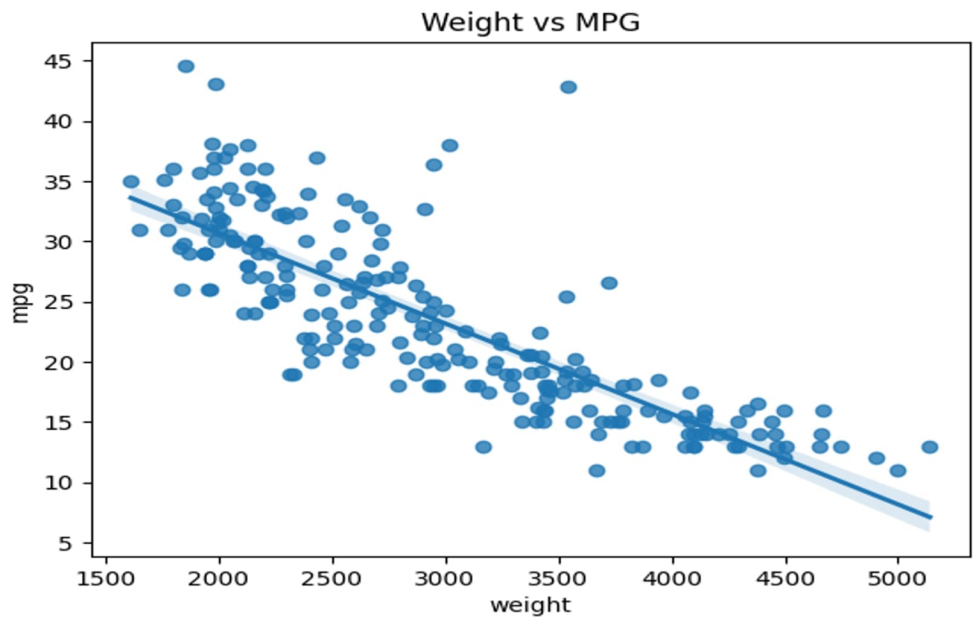


Figure: Scatter plot of Weight vs MPG with fitted regression line. Explanation: A simple linear regression model was fit with MPG as the response and Weight as the predictor. The notebook reported the fitted model as: $MPG = 45.7278632538935 + (-0.0075170047751075405) \times \text{Weight}$. The R^2 score reported was 0.677084005533519, indicating that approximately 67.7% of the variance in MPG is explained by vehicle weight. The negative slope confirms the inverse relationship: as weight increases, MPG decreases.

9. Conclusion

The matched report pairs the textual analysis from AutoMPG_Analysis_Report.pdf with the extracted plots from the notebook. Key findings: weight is strongly negatively associated with MPG; there are significant group differences in MPG by origin; and outliers/skewness in MPG suggest using robust statistics when appropriate. The notebook outputs (missing-value counts, descriptive statistics, test statistics, regression coefficients) were used to populate the explanations above.